

ALYANNA MARTINEZ

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EDUCATION

Bachelor of Arts in Computer Science, Psychology (Minor in Mathematics)

Expected: 05/2027

Southwestern University, Georgetown, TX

Honors: Dean's List, NSF S-STEM Scholarship, Mood Merit Scholarship, King Creativity Scholar, Sarofim Performance Award

Selected Coursework: Software Engineering, Machine Learning, Algorithms, Database Management, Programming Languages, Computer Organization, Computer Systems, Discrete Mathematics

SKILLS

Programming: C, C++, Java, Python, Haskell, Mercury, HTML, JavaScript, CSS, Assembly x86, Arduino.

Python Libraries: NetworkX, OpenCV, Seaborn, Plotly, pandas, Matplotlib, NumPy.

Languages: Spanish (advanced proficiency)

RESEARCH EXPERIENCE

Department of Computer Science, Supervisor: Arjun Chandrasekhar, Ph. D. | Southwestern University

Computational Biology Research Assistant

06/1/2026 – 07/23/2026

- Developed Python tools to analyze plant root architecture using graph-based representations of biological structures with NetworkX.
- Refactored a graph-processing pipeline from coordinate-based node identification to unique node identifiers to improve data integrity and reduce ambiguity during graph analysis.
- Implemented automated data-processing workflows to evaluate thousands of plant samples and generate experimental result datasets.
- Created data visualizations using Matplotlib, Seaborn, and Plotly to compare optimization methods and treatment conditions.
- Designed and optimized graph processing and traversal processes by implementing dynamic programming techniques to prevent unnecessary utilization of computational resources.

Department of Mathematics, Supervisor: Alison Marr, Ph.D. | Southwestern University

Independent Student Researcher

06/2024 – 08/2024

- Supported faculty-led research project exploring features of subtractive edge magic labeling graphs and applying graph theory concepts.
- Produced and evaluated graphs using established formulas and detected patterns informing the potential discovery of new theorems.
- Developed Python code using NetworkX to automate graph generation.

05/2025 – 08/2025

- Continued last summer's research project by expanding the Python code developed, enabling automated visualization of graphs for deeper analysis and pattern recognition.

EXPERIENCE

University Knowledge Assistant | *Senior Seminar in Software Engineering*

01/2026 – 05/2026

- Collaborated with a 4-person team to implement a full-stack architecture for a chatbot-inspired web application that answers user queries using Gemini API calls and data scraped from the Southwestern University website.
- Developed front-end and back-end components using FastAPI and htmx to connect user interface with backend operations.
- Participated in team planning and architecture discussions to align technical decisions and ensure team coordination.

Computer Science and Calculus Tutor

08/2025 – Present

Tutoring Center | Southwestern University, Georgetown, TX

- Provide one-on-one support to students in an engaging way to motivate student learning.
- Guide students through assignments to develop programming and mathematical problem-solving skills.
- Identify areas of growth and those in need of improvement to guide tailored feedback.